

# PLC I/O Configuration

## 1 I/O Incorporation Methods

### PLC I/O Classification

PLCs are primarily categorized based on how their Input/Output (I/O) section is integrated with the processor. There are two main configurations:

1. **Fixed I/O** (Integrated)
2. **Modular I/O** (Rack-based)

## 2 Fixed I/O Configuration

### Fixed I/O

Commonly found in small PLCs, the **Fixed I/O** configuration means the processor and the I/O terminals are packaged together in a single, non-removable unit.

#### Key Characteristics:

- **Integrated Package:** The entire controller comes in one housing with a fixed number of connections.
- **Cost Efficient:** Generally has a lower initial cost compared to modular systems.
- **Simplicity:** Easier to set up for small-scale applications.

### Limitations of Fixed I/O

- **Lack of Flexibility:** You are stuck with the predefined number and type of I/O points.
- **Serviceability Issues:** If any internal part (like a single output terminal) fails, the **entire unit** must typically be replaced.
- **Expansion Limits:** While some units allow adding expansion "bricks," the core capacity remains limited.

### 3 Modular I/O Configuration

#### Modular I/O

**Modular I/O** systems are divided into separate compartments or slots using a rack system. This allows for a "mix-and-match" approach where specific modules can be chosen based on the application's needs.

**A Modular Controller consists of:**

- **Rack (Chassis):** The physical frame that holds the modules.
- **Power Supply:** Often a separate module or built into the rack.
- **Processor Module (CPU):** Plugs into a dedicated slot.
- **I/O Modules:** Digital or analog cards that plug into various slots.
- **Operator Interface:** Used for programming and monitoring the system.

#### 3.1 The Backplane Concept

##### What is a Backplane?

The **Backplane** is a series of electrical contacts located at the rear of the rack.

- When a module is slid into a rack slot, it makes an electrical connection with the backplane.
- This backplane serves as the communication bus that connects the **PLC processor** to all I/O modules in the rack.

*The choice between fixed and modular I/O defines the scalability and maintenance path of an automation system.*